MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 3A4

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, power amplifier

Outline	---	5-2 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated filament

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	-f	a	g2	g1	g3 fct (-par)	a	+f

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Pp	Pg2	Ik	(B) Alt
Unit:	V	Vdc	Vdc	Vdc	W	W	mAdc	ft
Maximum:	1.6 or 3.2	165	---	150	2.2	1.0	27	See note 4
Minimum:	1.2 or 2.4	---	---	---	---	---	---	---
TEST CONDITIONS:	1.4	150	-8.4	90	---	---	---	---

GENERAL:

Qualification - Not required

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	See note 1	0.65	II	Ic1	0	-1.0	µAdc
1256	Electrode current (1) (anode)		0.65	II	Ib	9.2	17.4	mAdc
1256	Electrode current (2) (anode	Ef = 1.1 Vdc; (see notes 1 and 2)	0.65	II	Ib	16	---	mAdc
(B) 1246	Audio frequency noise	Esig = 0.75 Vac; Rp = 2,000 ohms	0.65	II	EB	---	17	vu
(B) 1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Rp = 2,000 ohms	---	---	Ep	---	500	mVac
1301	Filament current		---	---	If	180	220	mAdc
1256	Electrode current (screen)		---	---	Ic2	1.3	3.3	mAdc
1306	Transconductance		---	---	Sm	1,670	2,450	µmhos
1236	Power oscillation	F = 30 MHz; Eb = 150 Vdc; Ec2 = 135 Vdc; Ib = 20 mAdc; Ic1 = 0.13 mAdc; Rg1 = 0.2 Meg (see note 3)	---	---	Po	0.85	---	W
1341	Power output	Esig = 6.0 Vac; Rp = 8,000 ohms	---	---	Po	550	---	mW
1331	Direct-interelectrode capacitance	Without shield	---	---	Cg1p	---	0.35	pF
					Cin	3.9	5.7	pF
					Cout	3.0	5.4	pF
(B) 1211	Insulation of electrodes		---	---	---	---	---	---
(B) 1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group A	---	---	---	---	---	---
	Life-test end points (500 hours)	Transconductance or	---	---	Sm	1,310	---	µmhos
		Power output	---	---	Po	350	---	mW

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NOTES:

1. This test to be performed at the conclusion of the holding period.
2. Ebb = Ecc2 = 150 Vdc; Rp = Rg2 = 2,000 ohms; Rg1 = 220,000 ohms; Rg2 connected through 0.25 μF capacitor to ground; Esig = 15 Vac from a 500-ohm source through a 1.0 μF coupling capacitor. Immediately before this test the tube is to be operated for at least 3 minutes under conditions of, or equivalent to, the conditions of the test.
3. Useful power output.
4. Altitude test not required as operating voltage is less than 280 volts. Requirements covering altitude vs. voltage breakdown relationship are specified in basic document.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL, MI
Navy -
Air Force - 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2425-18)

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-E-1/108B, ELECTRON TUBE, RECEIVING TYPE 3AL

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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